IGCSE 0607 Practice Questions — E6.3

Circles, arcs and sectors

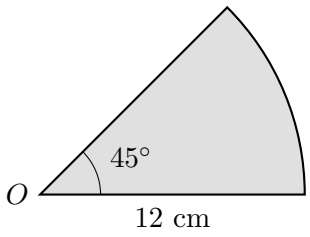
Total Questions: 9 — **Total Marks:** 60 Marks

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- **E6.3** Circles, arcs and sectors
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

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1 The diagram shows a sector of a circle, center O, with radius 12 cm and sector angle 45°.



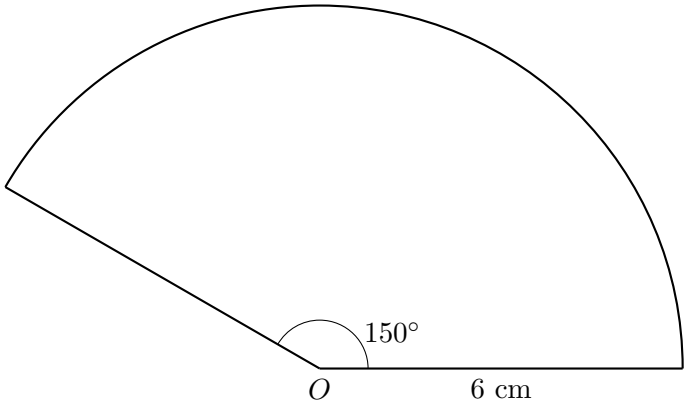
(a) Work out the exact length of the arc of the sector.
Give your answer in terms of π.

..... cm **[2]**

(b) Work out the exact area of the sector.
Give your answer in terms of π.

..... cm^2 **[2]**

2 The diagram shows a sector of a circle with centre O and radius 6 cm.
The sector angle is 150°.



(a) Calculate the area of the sector.
Give your answer in terms of π.

..... cm^2 **[2]**

(b) Calculate the perimeter of the sector.
Give your answer in terms of π.

..... cm **[3]**

3 **(a)** A sector of a circle, center O, has radius r cm and sector angle $\theta°$.
The area of the sector is 10π cm^2 and the length of its arc is 2π cm.

Find the value of r and the value of θ.

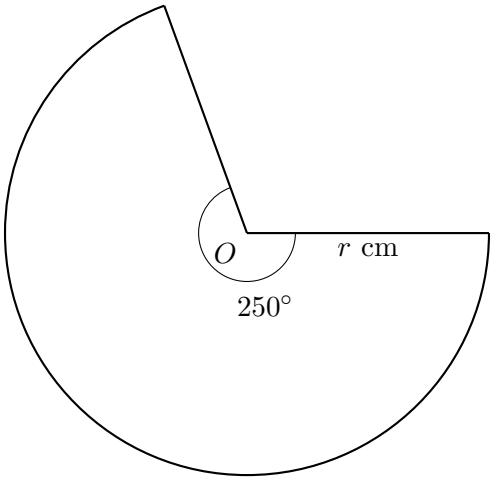
$r =$ ..... , $\theta =$ ..... **[5]**

(b) Another sector of a circle has radius x cm and sector angle $(x + 20)°$.
The length of its arc is $\frac{16\pi}{5}$ cm.

Find the value of x.
You must show all your working.

$x =$ ..... **[4]**

4 The diagram shows a major sector of a circle with centre O and radius r cm.
The reflex angle is 250°.
The length of the major arc is 35 cm.



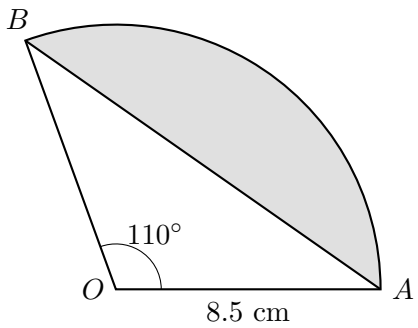
(a) Calculate the value of r.

$r =$ ..... **[3]**

(b) Calculate the area of the major sector.

..... cm^2 **[3]**

5 The diagram shows a sector OAB of a circle with center O and radius 8.5 cm.
The sector angle is 110°.
A straight line connects A and B, forming a triangle OAB and a shaded segment.



(a) Calculate the area of the sector OAB.

..... cm^2 **[2]**

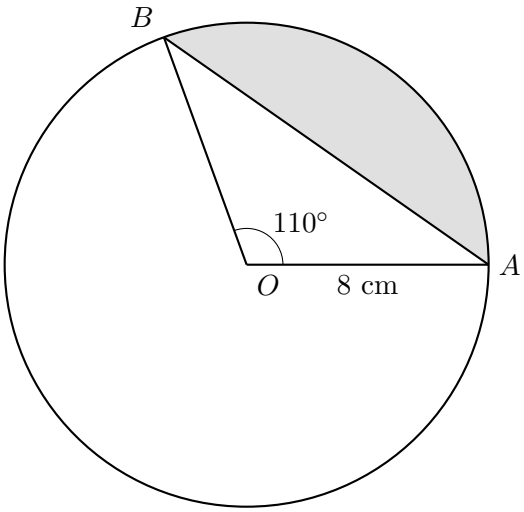
(b) Calculate the area of the shaded segment.

..... cm^2 **[3]**

(c) Calculate the perimeter of the shaded segment.

..... cm **[4]**

6 The diagram shows a circle with centre O and radius 8 cm.
A and B are points on the circumference such that angle $AOB = 110°$.



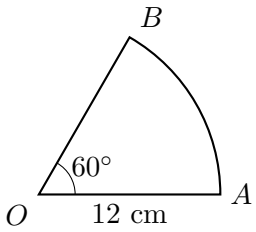
(a) Calculate the area of the shaded segment.

..... cm^2 **[4]**

(b) Calculate the perimeter of the shaded segment.

..... cm **[3]**

7 The diagram shows a minor sector AOB of a circle with centre O and radius 12 cm.
The sector angle is 60°.



NOT TO SCALE

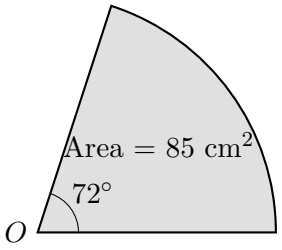
(a) Work out the length of the minor arc AB.
Give your answer in terms of π.

..... cm **[2]**

(b) Work out the area of the minor sector AOB.
Give your answer in terms of π.

..... cm^2 **[2]**

8 The diagram shows a sector of a circle with centre O and angle 72°.
The area of the sector is 85 cm^2.



NOT TO SCALE

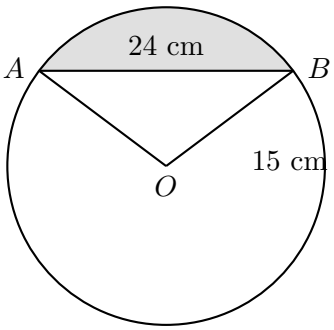
(a) Calculate the radius of the sector.

..... cm **[3]**

(b) Calculate the perimeter of the sector.

..... cm **[3]**

9 The diagram shows a circle, centre O, with radius 15 cm.
AB is a chord of the circle with length 24 cm.
The region bounded by the chord AB and the minor arc AB is shaded.



NOT TO SCALE

(a) Show that angle $AOB = 106.3°$, correct to 1 decimal place.

[3]

(b) For the following calculations, use angle $AOB = 106.3°$.

(i) Calculate the perimeter of the shaded minor segment.

..... cm **[3]**

(ii) Calculate the area of the shaded minor segment.

..... cm^2 **[4]**